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Paleomagnetism of the Cambrian Moore Hollow Group, Texas: Evidence for a Primary Magnetization Carried by Detrital Magnetite

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Petrographic and paleomagnetic studies of the Cambrian Moore Hollow Group, Texas, establish the presence of three magnetic minerals and several components of magnetization. Hematite is the dominant carrier of the magnetic remanence in most samples and is primarily a diagenetic replacement of preexisting Fe-bearing phases. Hematite-dominant samples have characteristic directions which range from shallow easterly to shallow southerly. This distribution suggests multiple components of magnetization acquired at various times in the Paleozoic. The paleopole obtained for an unmineralized breccia (135.7°E, 15.4°N) represents either a chemical remanent magnetization of Middle Ordovician to Early Silurian age or a vector addition of two or more Cambrian through Silurian components. Mineralized breccias at the Silver Creek Mississippi Valley-type deposit yield a Permian direction (117.0°E, 51.6°N). Because hematite in this deposit predates mineralization, the Permian age represents a maximum age for the mineralization at Silver Creek. Many samples contain goethite as a replacement of Fe-dolomite, glauconite, and pyrite. The direction of magnetization carried by goethite is near the present geomagnetic field or, in a minority of samples, antipodal to it. Rounded Ti- magnetite was identified by SEM in magnetic extracts from siliciclastic-poor facies. IRM acquisition and thermal demagnetization confirms the presence of magnetite in these samples. Some samples contain an apparent modern viscous remanent magnetization. The dominant component (72 samples, 10 sites) has a site mean declination of 93.8° and inclination of 14.5° with an $\alpha 95 = 8.5°$, corresponding to a pole position at 163.0°E, 0.6°S. The presence of detrital magnetite, polarity changes, lack of evidence for prolonged burial, and the proximity of this pole position to other late Precambrian/Cambrian poles suggest that this component of magnetization indeed reflects the Late Cambrian paleopole for cratonic North America.

Introduction

A widely accepted apparent polar wander path (APWP) for the pre-Carboniferous of stable North America remains a largely unattained goal. The major obstacle in establishing this portion of the APWP is the difficulty in distinguishing primary magnetizations from later Paleozoic remagnetizations. These secondary magnetizations include chemical remanent magnetizations (CRMs) resident in both magnetite [e.g. *McCabe et al.*, 1983, 1984; *Elmore et al.*, 1987; *Jackson et al.*, 1988] and hematite [*Ozdemir and Deutsch*, 1984; *Kent and Opdyke*, 1985; *Loucks and Elmore*, 1986; *Kent and Miller*, 1987; *Elmore et al.*, 1987], as well as thermoviscous remagnetization [*Kent*, 1985; *Jackson and Van der Voo*, 1986; *Moon and Merrill*, 1986] resident primarily in magnetite. Unequivocal primary magnetizations have been particularly difficult to demonstrate in Cambrian-aged rocks [e.g., *Seguin et al.*, 1981; *Lynnes and Van der Voo*, 1984].

We report here paleomagnetic results from the predominantly carbonate rocks of the Cambrian Moore Hollow Group in central Texas. Previous studies have demonstrated that hematite is the dominant magnetic carrier in this unit. *Van der Voo et al.* [1976] and Watts *et al.* [1980*a*] suggested that the hematite-resident component represents a primary magnetization, although they provided no supporting petrographic evidence. Loucks and Elmore [1986] recognized multiple hematite components in the Morgan Creek Limestone, a subunit of the Moore Hollow Group, and argued for a diagenetic origin for nearly all of the hematite based on petrographic observations. A small number of samples contained detrital grains that were composed of specular hematite, but it was unclear whether these grains were true detrital hematite or martite replacement of detrital magnetite.

Loucks and Elmore [1986] also documented the presence of magnetite in a few samples from the Morgan Creek Limestone. They suggested that the magnetite was probably detrital and should carry a detrital remanent magnetization (DRM). In this study we confirm the presence of a magnetization carried by magnetite in three subunits of the Moore Hollow Group and offer evidence that this magnetization is primary. We also establish the secondary nature of most of the hematite and goethite in the units that we sampled.

Geologic Setting

The Llano Uplift of central Texas exposes a Precambrian basement core rimmed by Cambrian through Cretaceous-aged sedimentary rocks (Figure 1). The Moore Hollow Group which occurs at the base of the section, includes two formations (Figure 2). The lower of these, the Riley Formation, has been subdivided into three members: the Hickory Sandstone, the Cap Mountain Limestone, and the Lion Mountain Sandstone [*Barnes and Bell*, 1977]. It was deposited on an eroded

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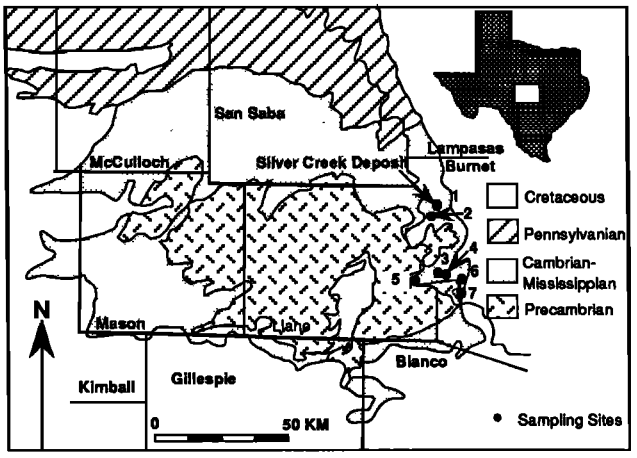


Fig. 1. Geologic map of the Llano area of central Texas showing the location of sampling sites. Site 1 includes the Silver Creek Mississippi Valley-type deposit.

Precambrian surface, part of which was emergent throughout its deposition; thus all members locally pinch out against Precambrian basement. The maximum thickness of the Riley Formation in the area we sampled was approximately 200 m. The Lion Mountain Sandstone, the oldest unit which we sampled, consists of glauconitic arenite (up to 80% glauconite), glauconitic siltstone, and interbeds and lenses of calcite-cemented echinoderm and trilobite grainstone which accumulated in a tidal inlet environment [*Chafetz*, 1978, 1979; *McBride*, 1988].

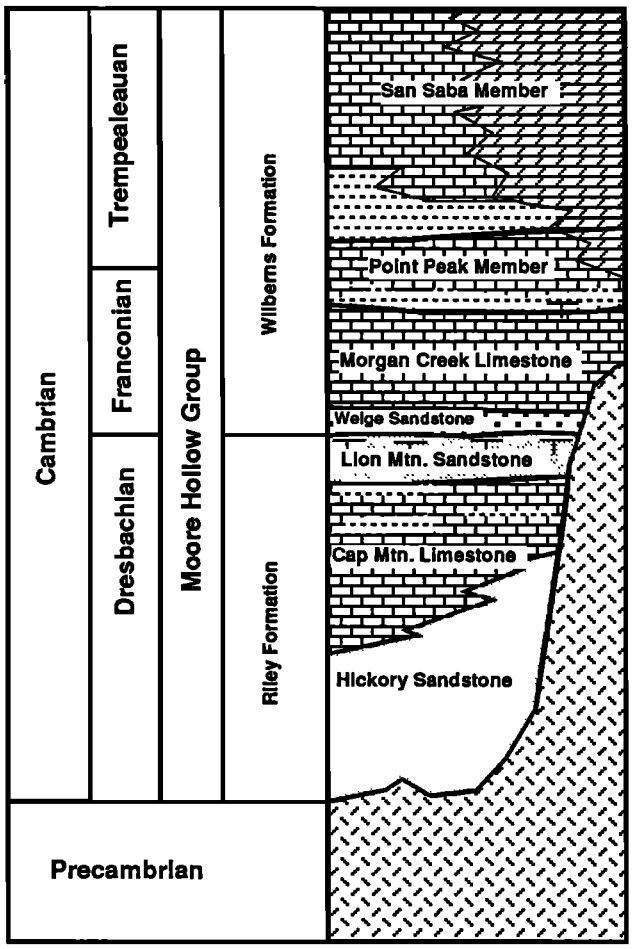


Fig. 2. Generalized stratigraphic column of the Cambrian rocks in the Llano region. Modified from *Barnes and Bell* [1977].

The Wilberns Formation which overlies the Riley Formation, contains four members: the Welge Sandstone, Morgan Creek Limestone, Point Peak Limestone, and San Saba Dolomite. In the sampling area the Wilberns attains a maximum thickness of approximately 250 m. The basal Welge Sandstone Member unconformably overlies the Riley Formation. It consists of a hematite-rich quartz arenite and represents shallow shelf deposition during a marine transgression. The Morgan Creek Limestone conformably overlies the Welge Sandstone and consists of a variety of facies deposited in a shallow shelf environment [*Barnes and Bell*, 1977]. The lower portion of the unit consists of hematitic sandy echinoderm grainstone which is overlain by thinly bedded trilobite packstone and medium-bedded ooid grainstone. It is capped by silty algal boundstone and/or wackestone. Conformably overlying the Morgan Creek Limestone is the Point Peak Member, of which we sampled the basal 10 m. It consists primarily of silty wackestone, siltstone, algal boundstone, and ooid packstone. The San Saba Dolomite was not sampled.

The Silver Creek deposit, a Mississippi Valley-type (MVT) lead-zinc deposit, occurs within small solution collapse structures of probable karst origin that penetrate much of the Upper Cambrian section (Figure 1, site 1) . Galena is the most common sulfide phase and occurs with associated pyrite, hematite, goethite, calcite, and dolomite within breccias and slumped beds. In an abandoned mine we took samples of clasts derived from the Lion Mountain Sandstone, Welge Sandstone, and Morgan Creek Limestone. Elsewhere in the deposit, mineralization occurs within beds of the Cap Mountain Limestone [*Barnes*, 1956].

PROCEDURE

The seven sampling sites are located along the northeastern edge of the Llano Uplift in western Burnet County (Figure 1). We collected 176 samples using a portable gasoline-powered drill. From these we cut 208 specimens. One hundred fifteen polished thin sections were examined with the petrographic microscope to identify opaque minerals, their relative abundances, and their paragenetic relationships. Coarser-crystalline hematite was primarily identified in thin section under reflected light. Finer-crystalline hematite and goethite were differentiated in thin section using the "blue card" technique of R. L. Folk (personal communication, 1985) in which a blue laminated card is inserted under the thin section when viewed in reflected light. The orange and red hematite and yellow goethite are accentuated using this technique. The examination of thin sections was augmented by detailed examination of 15 representative HCl-etched samples using a scanning electron microscope (SEM).

To aid in the identification of magnetic minerals, 10 samples were dissolved in a 4 m acetic acid solution buffered to a pH of 4.5 and the insoluble residue was passed over a hand magnet. The magnetic separate was then viewed using the SEM with an energy dispersive analyzer (EDA).

Throughout the paleomagnetic experiments, the samples were kept in a magnetically shielded room. During the course of the study an optimum demagnetization routine was devised whereby samples were heated to 150°C then subjected to

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stepwise AF demagnetization to approximately 50 mT, followed by stepwise thermal demagnetization to 700°C. This technique seemed to yield the best resolution of low blocking temperature, low-coercivity, and high blocking temperature components. A minimum of 10 demagnetization steps was used for all specimens. Remanences were measured using a two-axis ScT cryogenic magnetometer.

The acquisition of isothermal remanent magnetization (IRM) in fields up to 1.4 T was used to aid in identifying magnetic phases in 84 specimens. Demagnetization vectors were plotted on orthogonal vector diagrams and directions of components identified using principal component analyses [*Kirschvink*, 1980]. Best fit vectors with mean angular deviations greater than 12° were rejected.

PETROGRAPHIC RESULTS

Goethite was identified in thin section in most samples from both formations. It occurs in high concentrations only as an alteration product of ferroan dolomite (Figure 3*a*), pyrite, marcasite, or glauconite. Goethite is most abundant in the carbonate units, and it is very abundant within wackestone and packstone in the middle portion of the Morgan Creek Limestone as noted by *Loucks and Elmore* [1986]. It also occurs as a major constituent in the matrix of the mineralized breccia at the Silver Creek MVT deposit, where it apparently replaces pyrite, glauconite, and possibly marcasite (Figure 3*b*).

Hematite is very abundant in nearly all samples from the Lion Mountain, Welge, and lower Morgan Creek members and is identifiable in thin sections in most samples from all units. It occurs most frequently after glauconite (Figure 3*c*), pyrite, as a replacement of carbonate allochems (particularly echinoderm fragments (Figure 3*d*)), and as a cement or pore filling. We also observed rounded octahedral grains composed of hematite in the Lion Mountain, Welge, and lower Morgan Creek members which resemble the hematite grains described by *Loucks and Elmore* [1986]. The octahedral habit of these grains indicate that they are martitized magnetite grains.

Hematite in the Silver Creek MVT deposit occurs primarily

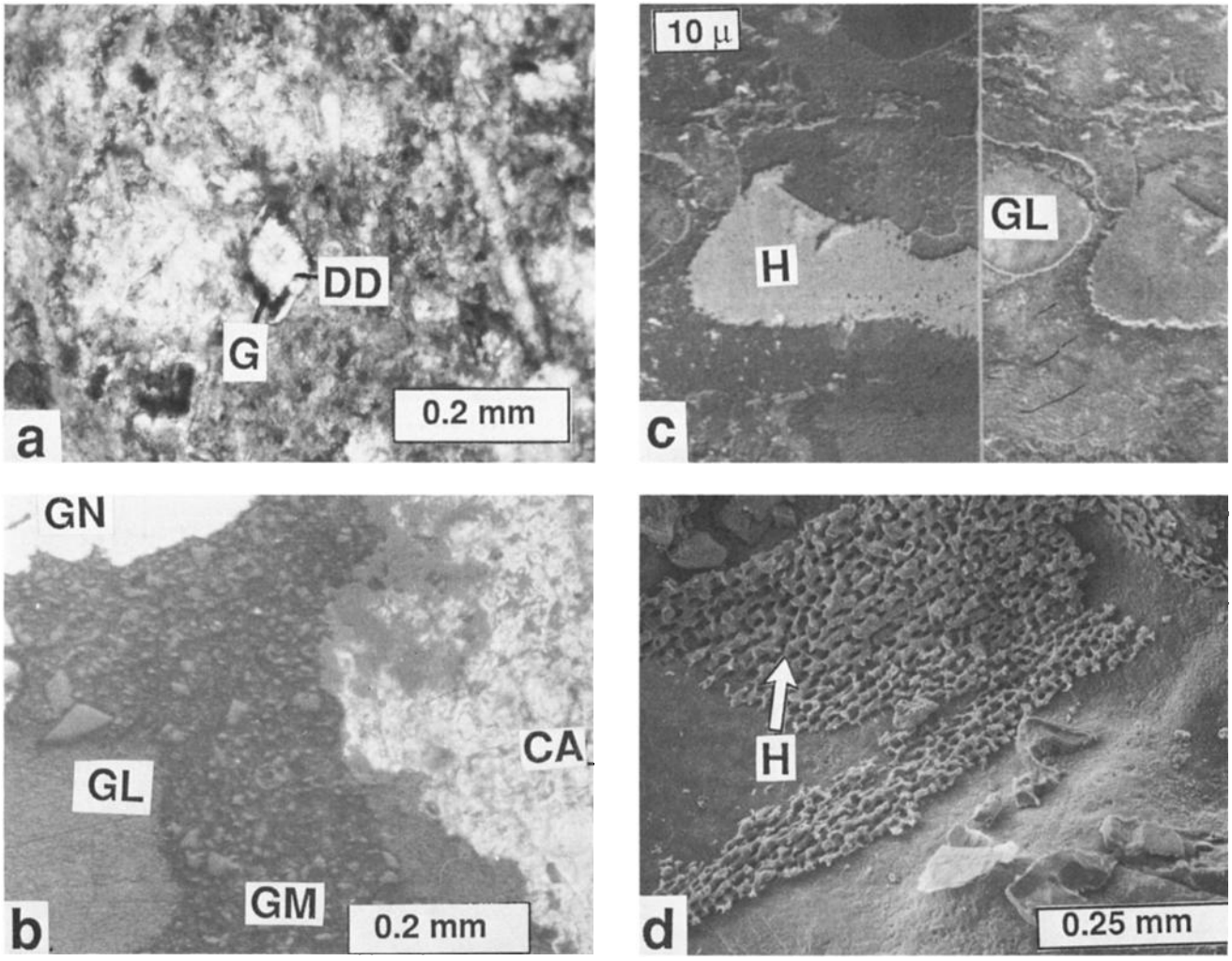


Fig. 3. (*a*) Photomicrograph of a Morgan Creek Limestone sample showing goethite (G) replacing ferroan zone of dedolomite (DD) (transmitted light). (*b*) Silver Creek sample in which goethite plus calcite makes up the matrix (GM) between breccia clasts composed of calcite (CAL), galena (GN), and glauconite (GL) (reflected and transmitted light). (*c*) Backscatter (left field of view) and SEM (right field of view) images of a sample from the Lion Mountain Member showing hematite (H) as a replacement of glauconite pelloids (GL). (*d*) SEM photo of hematite (H) replacing a portion of an echinoderm fragment in a sample from the Morgan Creek Limestone. (*e*) Hematite (H) in the Silver Creek Mississippi Valley-type deposit infills a fracture in a broken brachiopod fragment. Galena (GN) postdates the hematite (reflected light). (*f*) SEM photo of detrital magnetite grain in a sample from the Point Peak Limestone showing a possible exsolution lamellae of ilmenite (arrow).

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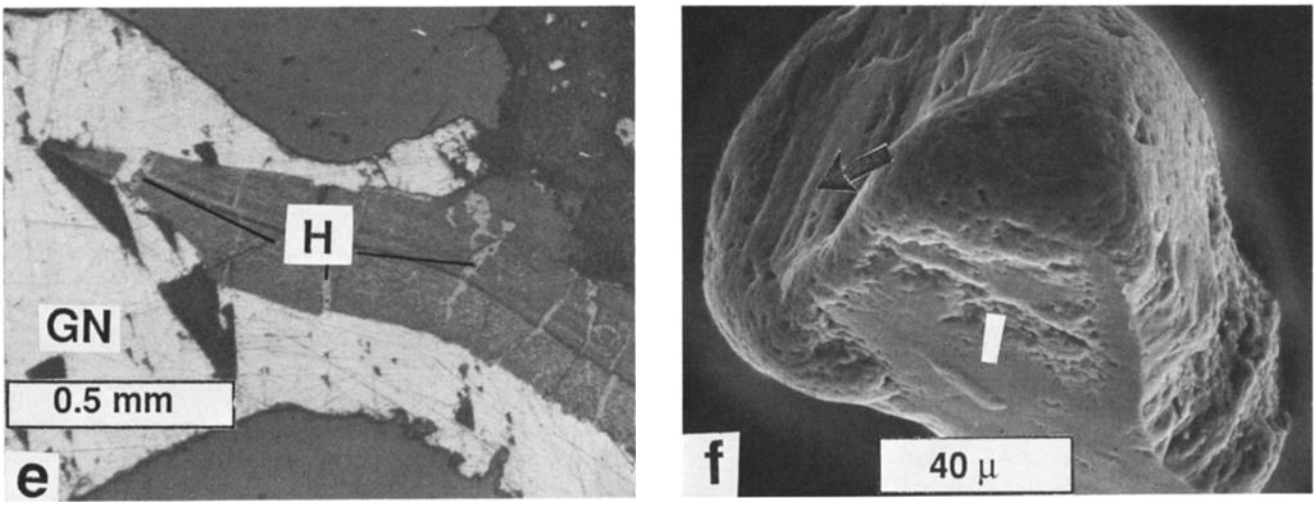


Fig. 3. (continued)

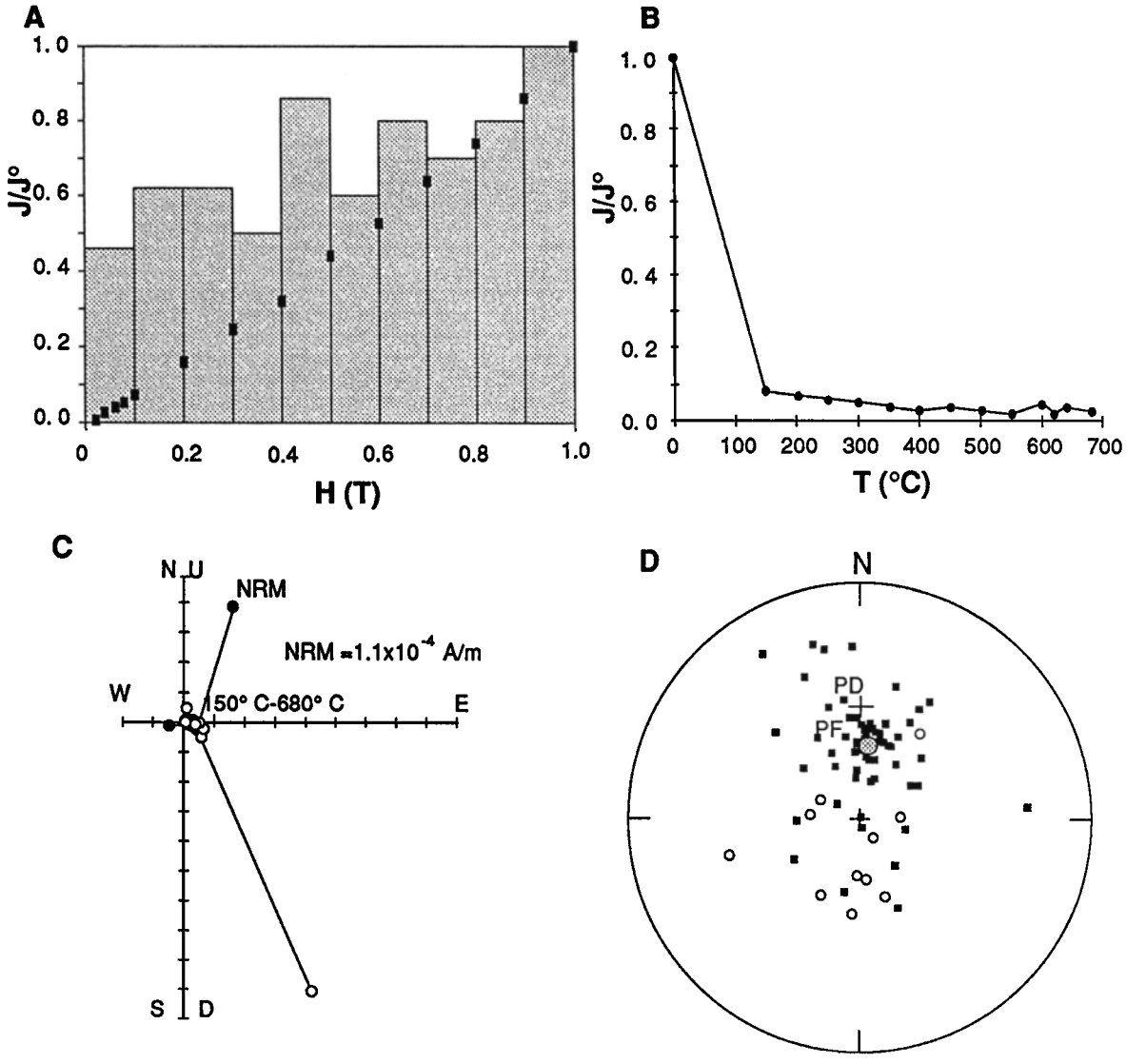


Fig. 4. Demagnetization and IRM behavior of an LT-dominant sample from the middle Morgan Creek Limestone. (*a*) Normalized IRM and coercivity spectra (histogram). Acquision of IRM indicates a dominant high-coercivity component which does not saturate in the maximum field of 1 T. (*b*) Thermal demagnetization removes nearly all of the NRM intensity by the first heating step. (*c*) Orthogonal thermal demagnetization diagram shows removal of a steep low blocking temperature component. Solid symbols represent projections onto horizontal plane; open symbols represent projections onto vertical plane. (*d*) Schmidt equal-area projection of low-temperature directions for LT samples. Solid (open) symbols denote positive (negative) inclination. Directions represent the vector erased between NRM and 150°C. All samples with negative inclinations are from sulfide-rich samples in the Silver Creek Mississippi Valley-type deposit. PD, present dipole field. PF, present field. Directions after structural correction.

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as a coarse-crystalline pore-filling and cement (Figure 3*e*). It predates sulfide mineralization, although the distribution of the coarse-crystalline hematite appears to be very closely associated with sulfide mineralization.

We did not recognize magnetite in thin section. However, magnetic mineral extracts from several hematite-poor samples yielded slightly rounded, equant grains having a maximum diameter of 80 μm. Many have an octahedral habit (Figure 3*f*). EDS analysis of the grains showed only Fe and Ti. Some of the grains showed striations, and the ratio of the intensity of Fe to Ti peaks fluctuated significantly in a traverse across the striations, indicating that these grains are detrital magnetite with exsolved ilmenite lamellae.

PALEOMAGNETIC RESULTS

Low Blocking Temperature (LT) Components

Low blocking temperature components occur in samples with abundant identifiable goethite. IRM saturation does not occur within the maximum applied fields of 1 T (Figure 4*a*). Most of the natural remanent magnetization (NRM) is removed during the first heating step of 150°C (Figures 4*b* and 4*c*). Most directions (Figure 4*d*) plot near present field (PF), which indicates that they were acquired during recent weathering. Nine directions are roughly antipodal to present field, indicating a pre-Brunhes age (>700,000 years BP). The samples with these directions are mineralized breccias from the Silver Creek MVT deposit (Figure 1) and apparently represent an earlier oxidation of pyrite and glauconite within the deposit.

High Blocking Temperature (HT) Components

Samples with high blocking temperature demagnetization behavior typically have abundant identifiable hematite in thin section. IRM curves of hematite-dominant samples indicated saturation at applied fields at or above 1.0 T (Figure 5*a*). The variation in saturating fields is a function of hematite crystal size, with coarser-crystalline samples saturating at lower

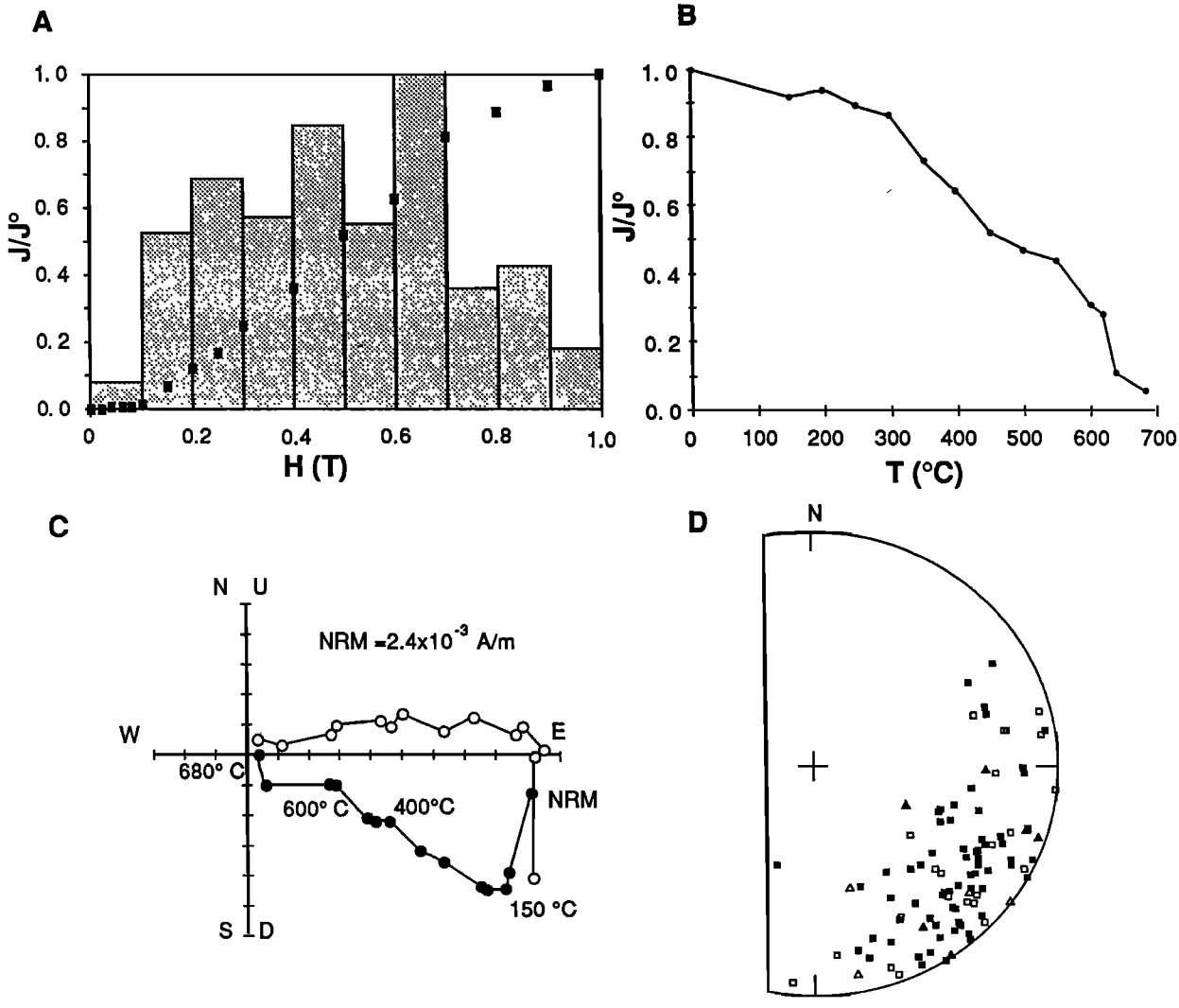


Fig. 5. IRM and demagnetization behavior of HT-dominant samples. (*a*) Acquisition of IRM indicates a dominant high-coercivity component which appears to approach saturation at about 1 T. (*b*) Demagnetization of NRM shows maximum unblocking temperatures in excess of 650°C. (*c*) Orthogonal thermal demagnetization diagram indicates the initial removal of a steep low blocking temperature component (compare Figure. 4*c*) followed by progressive removal of a shallow, southeasterly component. (*d*) Characteristic directions from HT-dominant samples exhibit a broad girdlelike distribution of shallow directions ranging from south to east. Solid symbols denote positive inclinations; triangles represent antipodals to northwesterly directions. Directions after structural correction.

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applied fields than fine-crystalline samples [*Dunlop*, 1972]. Thermal demagnetization of IRM indicates maximum unblocking temperatures ranging from 600° to 700°C (Figure 5*b*). Alternating field (AF) demagnetization to 90 mT was generally not effective in removing the NRM, although as much as 50% of the NRM intensity was removed from samples in which the hematite was very coarse crystalline. Principal component analyses of these samples yield a girdlelike distribution of directions, with shallow inclinations and declinations ranging from easterly to southerly (Figures 5*c* and 5*d*). A few antipodal northwesterly directions were isolated.

Although lack of significant bedding tilt prevented fold tests, we were able to perform conglomerate tests on hematite-dominant samples from two solution collapse breccias. Samples from a small unmineralized solution collapse breccia in the Morgan Creek Limestone at site 7 show a southeasterly cluster of characteristic directions with a mean declination of 124° and an inclination of 37° (Figure 6*a*). This corresponds to a virtual geomagnetic pole at 135.7°E longitude and 15.4°N latitude. Mineralized breccia samples from the Silver Creek MVT deposit containing coarse-crystalline hematite display more southerly directions with a mean of 159° and an inclination of 4° (Figure 6*b*) corresponding to a virtual geomagnetic pole at 117.0°E longitude and 51.6°N latitude. The grouping of directions from both breccias implies that magnetic components in both were acquired after breccia formation, i.e., the magnetizations are secondary. Bedding was not visible in the clasts; therefore we were unable to compare bedding-corrected directions from the breccias with uncorrected directions.

Low-Coercivity (LC) Components

Low-coercivity (LC) components include those in which a stable direction is defined over three or more demagnetization steps below 50 mT or below 580°C and which does not persist after heating above 580°C. Samples in which LC components dominate display IRM curves sith saturation at applied fields of less than 0.2 T (Figure 7*a*). The LC components are recognized only in samples with low abundances of hematite and were confirmed in all samples where we observed detrital magnetite in the magnetic separates. In some samples there is a minor contribution to the IRM curves by higher coercivity phases indicated by a steady rise above 0.2 T (Figure 8*a*). Maximum unblocking temperatures of the LC components between 500° and 575°C are indicated by demagnetization of NRM and IRM (Figure 7*b*, 8*b* and 8*c*). Orthogonal projections of demagnetization vectors indicate the dominance of shallow easterly directions in almost all samples, with a minority of samples displaying antipodal westerly directed directions (Figure 7*d*). This component shall be referred to as LC1. Some samples have an additional northerly directed component with steep positive inclinations (Figure 9*a*) which persists with AF demagnetization to at least 40 mT and with thermal demagnetization to 300°C. This component is interpreted as a modern viscous remanent magnetization (VRM) and shall be referred to as LC2.

Least squares fit vectors from LC components in which IRM curves indicate the presence of magnetite confirm the preponderance of shallow easterly directions (Figure 9*b*). Additional samples without accompanying IRM data, in which an intermediate blocking temperature and/or low-coercivity component was revealed by demagnetization, show a similar distribution of directions (Figure 9*c*). Five samples from carbonate grainstone lenses in the Lion Mountain Sandstone possess normal polarity (westerly directed) directions (plus symbols in Figures 9*b*, 9*c*) and 12 samples from all units display an apparent modern VRM component (LC2, Figure 9*a*).

LC-dominant samples are most abundant in the upper Morgan Creek and Point Peak members of the Wilberns Formation. They make up a minority of samples within grainstone lenses in the Lion Mountain Sandstone of the Riley Formation and are not present in sandstones of the Lion Mountain or Welge members and in sandy grainstones in the lower Morgan Creek Limestone where martite was identified petrographically.

The mean directions of the LC1 component for each site and unit are summarized in Table 1 and shown in Figure 10. The site means cluster well, giving a mean declination of 94° and an inclination of 15° with a cone of confidence at the 95% probability level of 8.5°. The corresponding virtual geomagnetic pole is located at 163.0°E longitude and 0.6°S latitude.

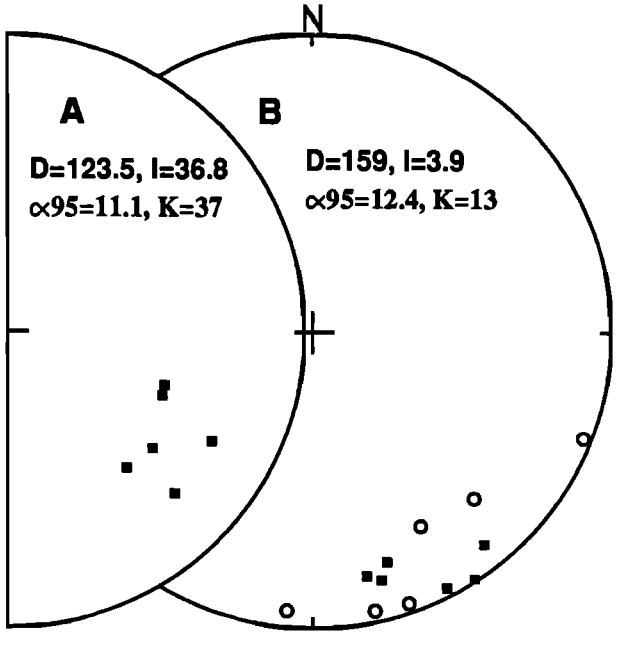


Fig. 6. Characteristic directions for samples from (*a*) nonmineralized solution collapse breccia in the Morgan Creek Limestone and from (*b*) mineralized breccia at the Silver Creek Mississippi Valley-type deposit. Solid (open) symbols denote positive (negative) inclination. No bedding correction applied.

DISCUSSION

High Blocking Temperature (HT) Components

Characteristic directions for high blocking temperature components (HT) clearly have a girdlelike and not a Fisherian distribution (Figure 5*d*). They represent either multiple magnetic components or vector additions of two or more hematite-resident components with overlapping blocking temperature spectra. As previously discussed, almost all of the hematite observed in thin section has a diagenetic origin. It is possible that some of the martite in the lower three units formed prior to deposition. However, the restriction of the martite to sandier portions of the section where initial

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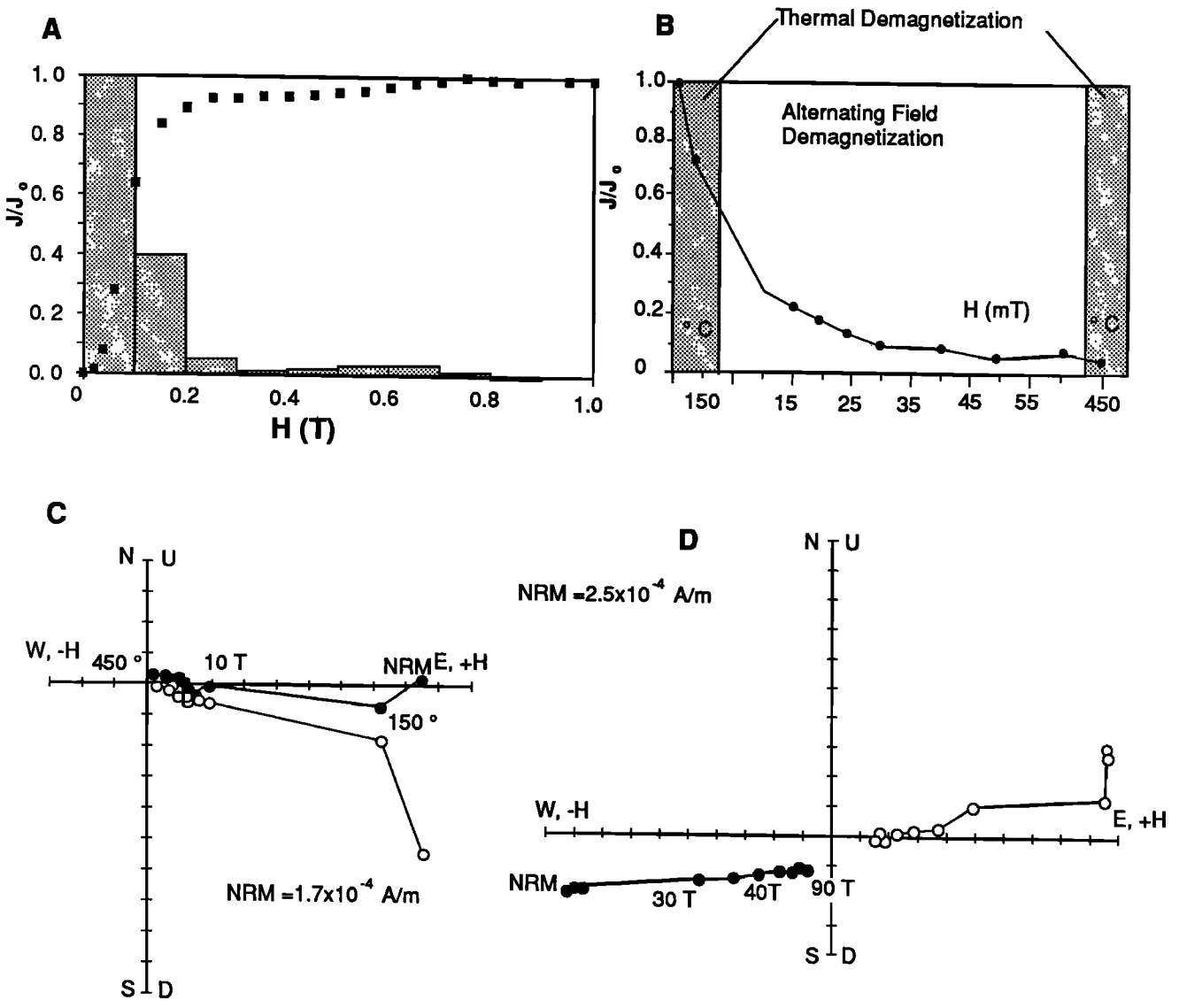


Fig. 7. IRM and demagnetization behavior of LC-dominated samples. (*a*) Acquisition of IRM from a pelletal wackestone sample from the lower Point Peak Member indicates a dominant low-coercivity component which saturates below 0.2 T. (*b*) Combination thermal and AF demagnetization of NRM. (*c*) Orthogonal demagnetization diagram shows initial removal of steep low blocking temperature component followed by progressive removal of a shallow easterly component with further AF demagnetization. (*d*) Orthogonal AF demagnetization of a trilobite grainstone from the Lion Mountain Sandstone indicates the removal of a shallow westerly component which is antipodal to directions shown in Figure 7*c*. Directions after structural correction.

permeabilities were high, and the presence of nonmartitized magnetite grains in the denser carbonates elsewhere in the section, suggest that the martite formed after deposition. A secondary origin for the hematite was also clearly established by the conglomerate tests for the two breccias (Figure 6).

The time of hematization can be estimated by comparing the pole position obtained from the breccia samples with the North American polar wander path (APWP after *Van der Voo* [1990]). The paleopole from the breccia in the unmineralized Morgan Creek Limestone (B1 in Figure 11) falls near the Middle Ordovician to Lower Silurian Pole from *Van der Voo* [1990]. The magnetization may reflect a single Late Ordovician to Early Silurian CRM. Alternatively, the magnetization could represent the vector addition of two or more Cambrian through Silurian CRMs. Either interpretation is supported by the work of *Morton and Long* [1984] and *McBride* [1988] who suggested that the first prolonged period of subaerial emergence and the first opportunity for extensive meteoric diagenesis in the Moore Hollow Group in this region occurred between the Late Ordovician and Devonian.

The pole position from mineralized breccias at the Silver Creek MVT deposit (B2 in Figure 11) lies between the Early and Late Permian poles on the APWP. The indicated Permian age apparently represents a maximum age of mineralization and a minimum age of brecciation, because hematite is older than galena but younger than the brecciation. However, the fact that this hematite is anomalously coarse crystalline, occurs predominantly as a cement, and is closely associated with MVT mineralization suggests that it may be an early precipitated phase in the MVT mineralizing event.

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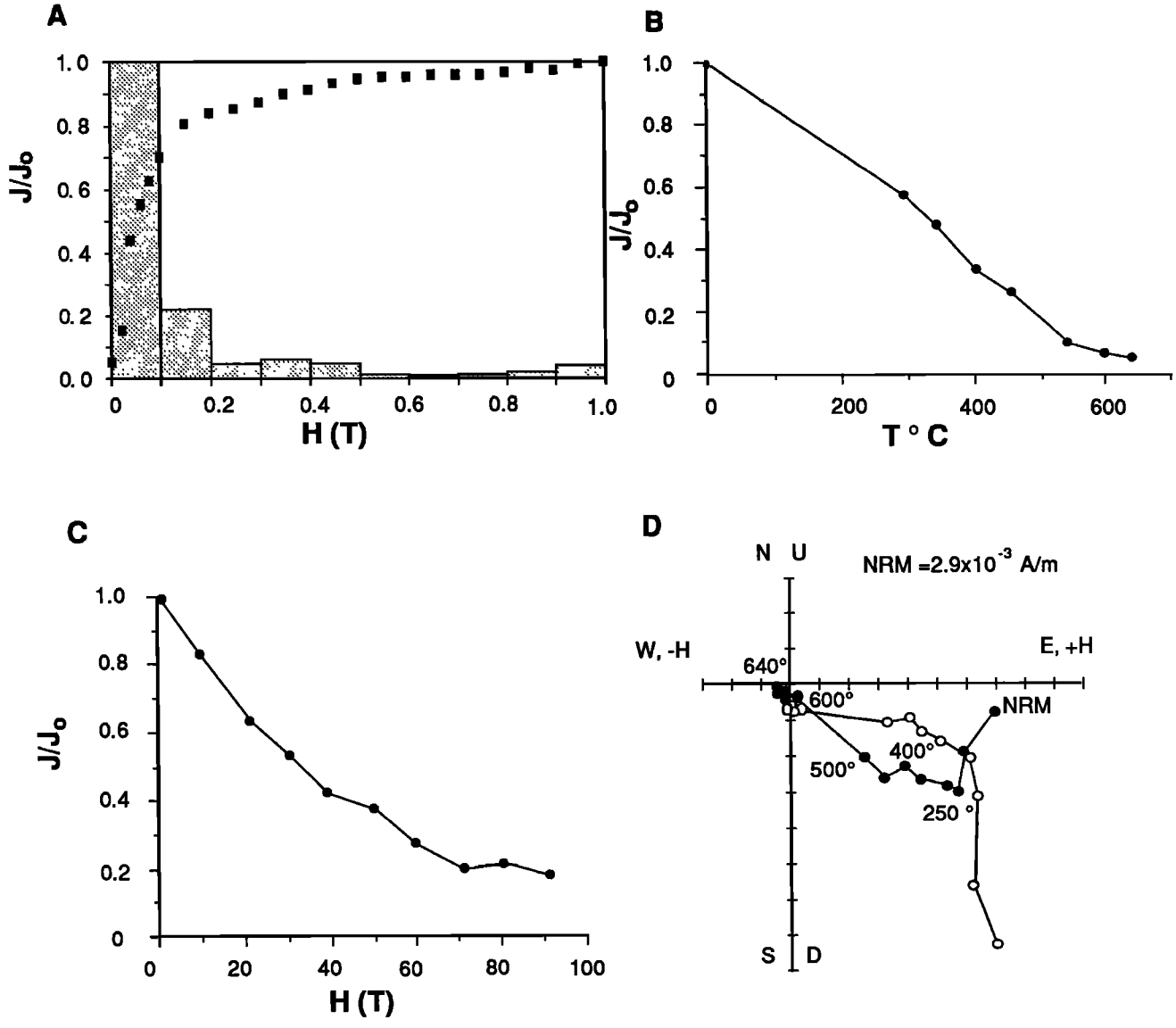
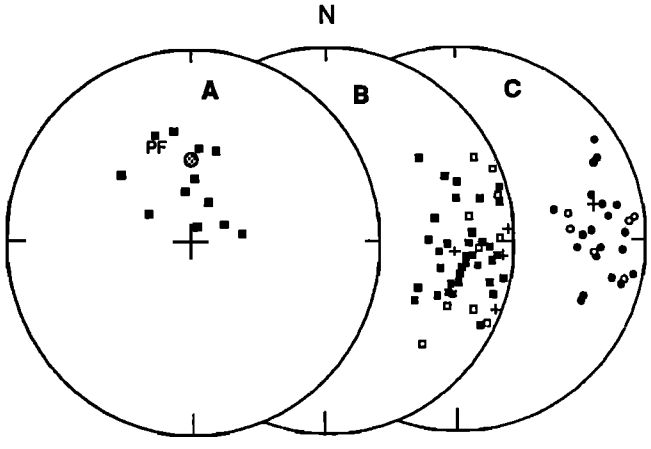


Fig. 8. Demagnetization and IRM behavior of echinoderm grainstone sample from the middle portion of the Morgan Creek Member. (*a*) Acquision of IRM indicates a dominant low-coercivity component which saturates below 0.2 T, and a higher-coercivity component which saturates above 0.5 T. (*b*) Stepwise thermal demagnetization of IRM shows removal of almost all of IRM by 550°C. (*c*) Demagnetization of NRM. Most of NRM is removed by 70 mT. (*d*) Orthogonal thermal demagnetization diagram indicates progressive removal of a shallow easterly component. From 600° to 640°C, there is progressive removal of a small shallow, southeasterly oriented component. Directions after structural correction.



We postulate that the hematite formed from Fe^{2+} - enriched basinal fluids as they encountered redox barriers within the permeable solution collapse breccias. These fluids were probably expelled from a sedimentary basin along the Ouachita-Marathon fold and thrust belt to the east during late Paleozoic tectonic activity. MVT minerals may have formed later during this same fluid migration event.

Fig. 9. Characteristic directions from samples with low-coercivity characteristics (LC component). (*a*) Low stability directions group near present field (PF). (*b*) Samples in which IRM curves indicated the presence of magnetite. (*c*) Additional low-coercivity samples for which IRM curves were not measured. Crosses represent antipodal direction of normally magnetized samples. Directions after structural correction.

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TABLE 1. Site Mean Directions and Paleopoles for Component LC1

Site	Stratigraphic Unit	Mean Declination	Mean Inclination	K	$\propto$ 95	N/N'/NT
1	LM	94.7	-6.0	6.5	25.8	7/10/15
1	MC	93.4	5.5	18.0	22.3	4/5/5
1	BX	------	------	-----	-----	0/0/12
1	W	------	-----	-----	-----	0/0/2
2	LM	90.8	-0.6	5.0	38.1	6/7/28
2	W	------	-----	-----	-----	0/0/3
2	MC	99.0	19.5	27.2	11.8	7/8/8
3	PP	90.6	25.7	6.1	14.4	20/27/27
4	PP	87.2	30.3			2/5/5
5	LM	96.3	26.5	33.8	16.0	4/7/53
6	MC	102.3	24.1	84.6	10.0	5/7/7
6	PP	87.9	19.5	7.9	14.5	13/16/16
7	MC	95.9	0.2	9.2	26.6	5/5/19
7	W	------	-----	-----	-----	0/0/7
All sites		93.8	14.5	33.2	8.5	10

Paleolatitude	Paleolongitude	δp	δm
0.6°S	163.0°E	4.5	8.7

N is the number of directions determined. N' is the total number of samples with LC characteristics. NT is the total number of samples. All directions are bedding corrected. Stratigraphic units are LM, Lion Mountain; MC, Morgan Creek; PP, Point Peak; and W,Welge. BX is mineralized solution collapse breccia.

Similar paleomagnetic ages have been obtained from MVT deposits and remagnetized lower Paleozoic strata from many places on the North American midcontinent [e.g. *Wisniowiecki et al.*, 1983; *McCabe et al.*, 1983; *Bachtdase et al.*, 1987] and have been used as evidence for a continent-wide fluid migration event during this time [*Oliver*, 1986]. Although most of the Kiaman remagnetization has been attributed to authigenic magnetite, preliminary results from the Ordovician Viola Formation in Oklahoma seem to confirm a genetic relationship between MVT mineralization and Permian-aged coarse-crystalline authigenic hematite [*Elmore et al.*, 1989], very similar to what we observed in this study.

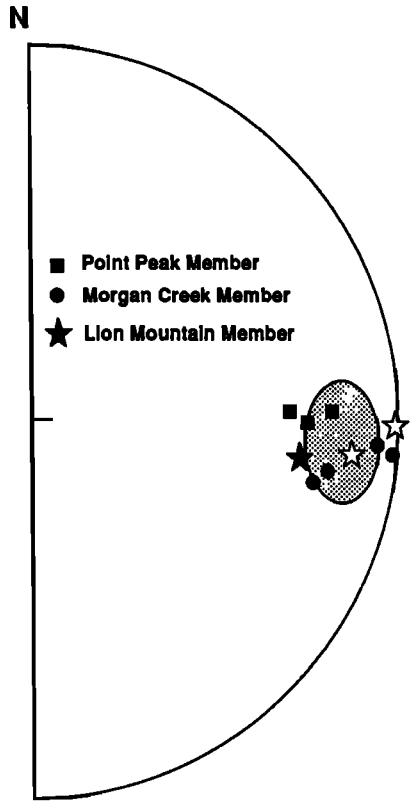


Fig. 10. Site means and 95% cone of confidence for Lion Mountain, Morgan Creek, and Point Peak members. Solid symbols denote positive inclinations. Note the similarity in directions between the three units. Directions after structural correction.

Low-Coercivity (LC) Components

IRM acquisition and the demagnetization behavior as well as the SEM results strongly suggest that magnetite is the carrier of both LC components. The slightly rounded

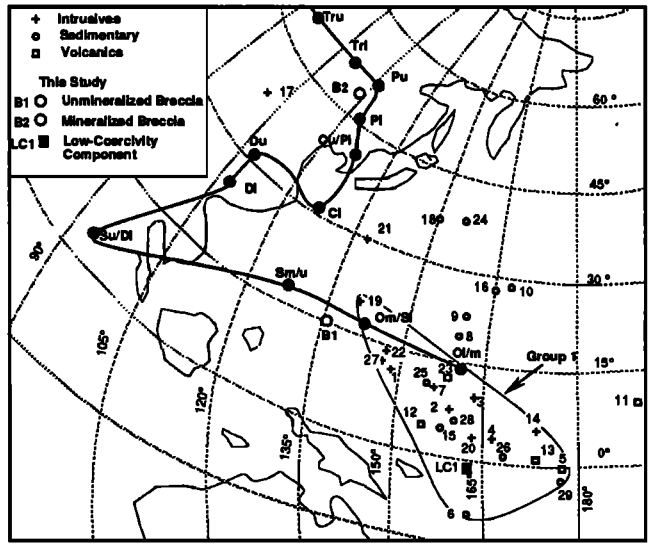


Fig. 11. North American apparent polar wander path [after *Van der Voo*, 1990] and published latest Precambrian and Cambrian pole positions, coded to Table 2. Large square shows the pole position of the LC component of this study, and circles represent the poles of the unmineralized breccia form the Morgan Creek Limestone (B1) and the mineralized breccia from the Silver Creek deposit (B2).

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appearance of the magnetite grains certainly suggests a detrital origin, as do the apparent exsolution lamellae of ilmenite within them. By contrast, authigenic magnetite in Paleozoic rocks is typically spherical and contains little or no titanium [*McCabe et al.*, 1987]. The magnetite grains which we have identified under SEM are clearly too large (>30 μm in diameter) to be the actual carriers of the LC component. However, their identification supports the likely presence of smaller magnetite particles capable of carrying the magnetization.

In addition to the SEM evidence for a detrital origin for the LC component, the Moore Hollow Group data base meets all of the criteria for a primary magnetization as prescribed by *Van der Voo* [1988; 1990] with the exception of field tests which would help constrain the timing of magnetization relative to folding or brecciation (criterion 4 [Van der Voo, 1990]). Bedding corrections were always less than 5°; thus bedding-corrected and uncorrected directions were essentially identical. As previously discussed, the conglomerate tests on two breccias from the Moore Hollow Group indicated a postbrecciation magnetization resident in hematite. We found no indication of the low-coercivity components in these samples.

The presence of both reversed and normal polarity intervals (criterion 6 [*Van der Voo*, 1990]) in LC1 samples is consistent with a primary magnetization, although it is not diagnostic. Secondary magnetizations can span multiple reversals and preferentially affect specific beds at different times, thus producing a "false" magnetostratigraphy. However, the presence of reversals does indicate that enough time elapsed during the acquisition of magnetization to preclude instantaneous remagnetization [*Van der Voo*, 1990]. There was inadequate stratigraphic resolution in our study to determine a consistent magnetostratigraphy, although within a given site, most reversals were defined by more than one specimen.

To our knowledge, there are no published independent estimates of burial temperatures for the area such as conodont alteration indices (CAI). However, the Moore Hollow Group in the area probably never experienced more than 1 km of burial based on maximum existing thicknesses of all stratigraphic units in the region [*McBride*, 1988]. Therefore a thermoviscous origin of component LC1 acquired during burial [e.g., *Tucker and Kent*, 1988] is unlikely.

A primary origin for LC1 is also suggested by the proximity of its pole position to older paleopoles and its lack of similarity to younger paleopoles (criterion 7 [Van der Voo, 1990]). Figure 11 and Table 2 summarize the most frequently cited paleopoles for the Cambrian and latest Precambrian of North America. Twenty-one out of the 29 pole positions, including all six late Precambrian to Early Cambrian poles, form a distinct grouping (the area outlined in the Figure 11, group 1). The remaining pole positions scatter broadly around the Paleozoic APWP which strongly suggests remagnetization. Although *Van der Voo* [1990] shows an overall northwestward shift of Cambrian poles with decreasing age, there is tremendous overlap of circles of confidence for these poles. An examination of Figure 11 indicates little or no consistent trend in the distribution of Lower, Middle, and Upper Cambrian poles. The observed scatter could be due to secular variation, postdepositional magnetization, partial remagnetization or small-scale tectonic motions. It seems significant that the latest Precambrian poles, with the exception of pole 8 from the Charleroix Anorthosite, Quebec [*Hargraves and Roy*, 1974], fall into the southeastern half of the delineated area.

There is evidence that many of the poles in the western portion of this group of Cambrian poles may represent secondary magnetizations. In spite of having a quality criteria (Q [*Van der Voo*, 1988, 1990]) value of greater than 3, five of these poles are from hematite components in sedimentary rocks, and the difficulty in establishing the time of magnetization in hematitic sediments is well documented [e.g,. *Larson and Walker*, 1975; *Turner and Archer*, 1975]. In particular, the Tapeats Formation (pole 15 [*Elston and Bressler*, 1977]) was given a Q value of 5 by *Van der Voo* [1990]. Nevertheless, petrographic evidence from *Gillett* [1982] indicates that this pole represents a secondary component residing in diagenetic hematite.

The northwest-southeast trend of group 1 poles therefore appears to be largely a result of vector additions of two or more Cambrian and post-Cambrian magnetizations. The pole position derived from the LC1 component in this study falls within the southeastern portion of group 1. Given the evidence for a primary origin for LC1 and the lack of compelling evidence for a primary or early diagenetic origin for many of the poles in the northwestern portion of the delineated area, we suggest that pole LC1 represents the best estimate for a Late Cambrian pole position.

The Cambrian Loop

Earlier paleomagnetic results from the Moore Hollow Group [*Van der Voo et al.*, 1976; *Watts et al.*, 1980a, Table 2], in conjunction with other published data, led these authors to propose the existence of a "Cambrian loop", in which the virtual geomagnetic pole migrated from a position near 165°E, 0°N in the Early Cambrian to 95°E, 45°N for the Middle Cambrian, then back to 160°E, 6°N in the Late Cambrian. The validity of the Cambrian loop has been questioned by several later workers [*Lynnes and Van der Voo*, 1984; *Loucks and Elmore*, 1986; *Dunn and Elmore*, 1985; *Larson et al.*, 1985] because of the lack of evidence for a primary origin of the magnetization as well as the amount and directions of the implied plate motions. Our results from the LC1 component (see Figure 11) differ significantly from these previous results and, most importantly, imply no dramatic excursions in the Cambrian APWP. In light of our petrographic evidence for a secondary origin of the hematite in some of the same rock units sampled by *Van der Voo et al.* [1976] and *Watts et al.* [1980*a*], it is more reasonable to interpret their pole positions as a combinations of two or more secondary components spanning much of the Paleozoic. Parenthetically it might be added that this latter interpretation was favored by one of Watt's coauthors in his thesis [*Reeve*, 1975].

SUMMARY AND CONCLUSIONS

On the basis of petrographic identification, demagnetization behavior, and coercivity spectrum analyses, we identified three magnetic phases in the Moore Hollow Group. Hematite is the dominant carrier of magnetization in most samples and occurs primarily as a diagenetic replacement mineral. Directions from hematite-dominant samples probably represent discrete post-Cambrian CRMs as well as superposition of two or more post-Cambrian CRMs.

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TABLE 2. Previously Published Cambrian and Late Precambrian Paleopoles From North America

No.	Units, Location	Age	Longitude °E	Latitude °N	Principal Carrier	Reference
1	Charleroix Anorthosite, Quebec	L. Precambrian	148	12	Magnetite?	*Hargraves and Roy* [1974]
2	Indian Head Anorthosite, Quebec	L. Precambrian	158	8	Magnetite?	*Murthy and Rao* [1976]
3	Morin Anorthosite, Quebec	L. Precambrian	164	9	Magnetite?	*Irving and McGlynn* [1976
4	Franklin intrusives, Northwest Territories	L. Precambrian - Cambrian	166	4	Magnetite?	*Fahrig et al.* [1971]
5	Unicoi Basalts, Virginia	L. Precambrian - Cambrian	178	0	Hematite	*Brown and Van der Voo* [1982]
6	Johnnie Rainstorm, Nevada	L. Precambrian - Cambrian	162	10°S	Hematite	*Gillett and Van Alstine* [1979]
7	Intrusives, Colorado	Cambrian	156	12	Magnetite?	*French et al.* [1977]
8	Bradore Sandstone, Quebec	E. Cambrian	159	19	Hematite	*Rao and Deutsch* [1976]
9	Bradore Sandstone, Quebec	E. Cambrian	160	23	Hematite	*Rao and Deutsch* [1976]
10	Bradore Sandstone, Quebec	E. Cambrian	167	29	Hematite	*Rao and Deutsch* [1976]
11	Buckingham volcanics, Quebec	E. Cambrian	186	10	Magnetite?	*Dankers and Lapointe* [1981]
12	Buckingham volcanics, Quebec	E. Cambrian	154	6	Magnetite?	*Dankers and Lapointe* [1981]
13	Buckingham volcanics, Quebec	E. Cambrian	173	1	Magnetite	*Dankers and Lapointe* [1981]
14	Cloud Mountain Basalt, Newfoundland	E. Cambrian	173	5	Magnetite	*Deutsch and Rao* [1977]
15	Tapeats Formation, Arizona	E.to M. Cambrian	158	5	Hematite	*Elston and Bressler* [1977]
16	Waynesboro Formation, Appalachians	E.to M. Cambrian	165	28	Hematite	*Watts et al.* [1980a]
17	Intrusives, Colorado	M. Cambrian	101	37	Magnetite?	*Larson et al.* [1985]
18	Rome Formation, Appalachians	M. Cambrian	150	36	Hematite	*Watts et al.*,[1980b]
19	Sept-Iles Intrusion, Quebec	M. Cambrian	141	20	Magnetite?	*Tanczyk et al.* [1987]
20	Wichita Granite, Oklahoma	M. Cambrian	164	4	Magnetite	*Spall* [1971]
21	Wichita Granite, Oklahoma	M. Cambrian	138	30	Magnetite	*Vincenz et al.* [1975]
22	McClure Mtn. complex, Colorado	M. to L. Cambrian	146	14	Magnetite	*Lynnes and Van der Voo* [1984]
23	Volcanics, Quebec	M. to L. Cambrian	157	13	Magnetite	*Seguin et al.* [1978]
24	Chateauguay Member, Quebec	L. Cambrian	156	37	Magnetite	*Sequin et al.* [1981]
25	Lamotte Sandstone, Missouri	L. Cambrian	155	12	Hematite	*Al-Khafaji and Vincenz* [1971]
26	Lamotte Sandstone, Missouri	L. Cambrian	168	1	Hematite	*Van der Voo et al.* [1976]
27	Ophiolite, Quebec	L. Cambrian	146	13	Magnetite?	*Seguin et al.* [1978]
28	Point Peak Member, Texas	L. Cambrian	159	6	Hematite	*Van der Voo et al.* [1976]
29	Taum Sauk Limestone	L. Cambrian	176	4°S	Hematite	*Dunn and Elmore* [1985]
	Moore Hollow Group, Texas					
	Hickory	M. toL. Cambrian	145	34	Hematite	*Watts et al.* [1980a]
	Cap Mountain	L. Cambrian	140	33	Hematite	*Watts et al.* [1980a]
	Lion Mountain	L. Cambrian	146	27	Hematite	*Watts et al.* [1980a]
	Welge-Morgan Creek	L. Cambrian	151	24	Hematite	*Watts et al.* [1980a]
	Morgan Creek Limestone	L. Cambrian	158	11	Hematite	*Loucks and Elmore* [1986]
	Point Peak	L. Cambrian	159	6	Hematite +Magnetite	*Van der Voo et al.* 1976]

Numbers refer to pole numbers in Figure 11. E. is early, M is middle and L is late.

Hematite in a mineralized breccia at the Silver Creek MVT deposit yields a Permian paleopole which indicates a maximum age for mineralization. However, the close spatial association of the hematite with MVT mineralization as well as the textural evidence that the hematite may have formed from high temperature, basin-derived fluids suggest that hematite and mineralization may be related. It is postulated that both hematite and MVT sulfide minerals formed from ascending basinal fluids which were expelled from a sedimentary basin along the Ouachita-Marathon fold and thrust belt during late Paleozoic tectonic activity.

Hematite from an unmineralized breccia in the Morgan Creek Limestone yields a paleopole which falls near the Middle Ordovician to Lower Silurian portion of the North American APWP. It probably formed during meteoric alteration associated with a major drop in sea level after the Middle Ordovician [e.g., *Morton and Long*, 1984]. Goethite in the Moore Hollow Group is the result of recent weathering, dating back to pre-Brunhes times in the case of the Silver Creek MVT deposit.

A magnetization carried by magnetite is present in the carbonate-rich portions of the Moore Hollow Group. Textural and geochemical evidence revealed that the magnetite is of detrital origin. The presence of polarity reversals and the close agreement with published late Precambrian and Cambrian pole positions are further arguments for a primary magnetization. The possibility of a thermoviscous magnetization can be ruled out because of insufficient burial depth or duration. Thus the magnetite-residing magnetization indeed reflects the Late Cambrian field direction which corresponds to a virtual geomagnetic pole position for cratonic North America at 163.0°E, 0.6°S.

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